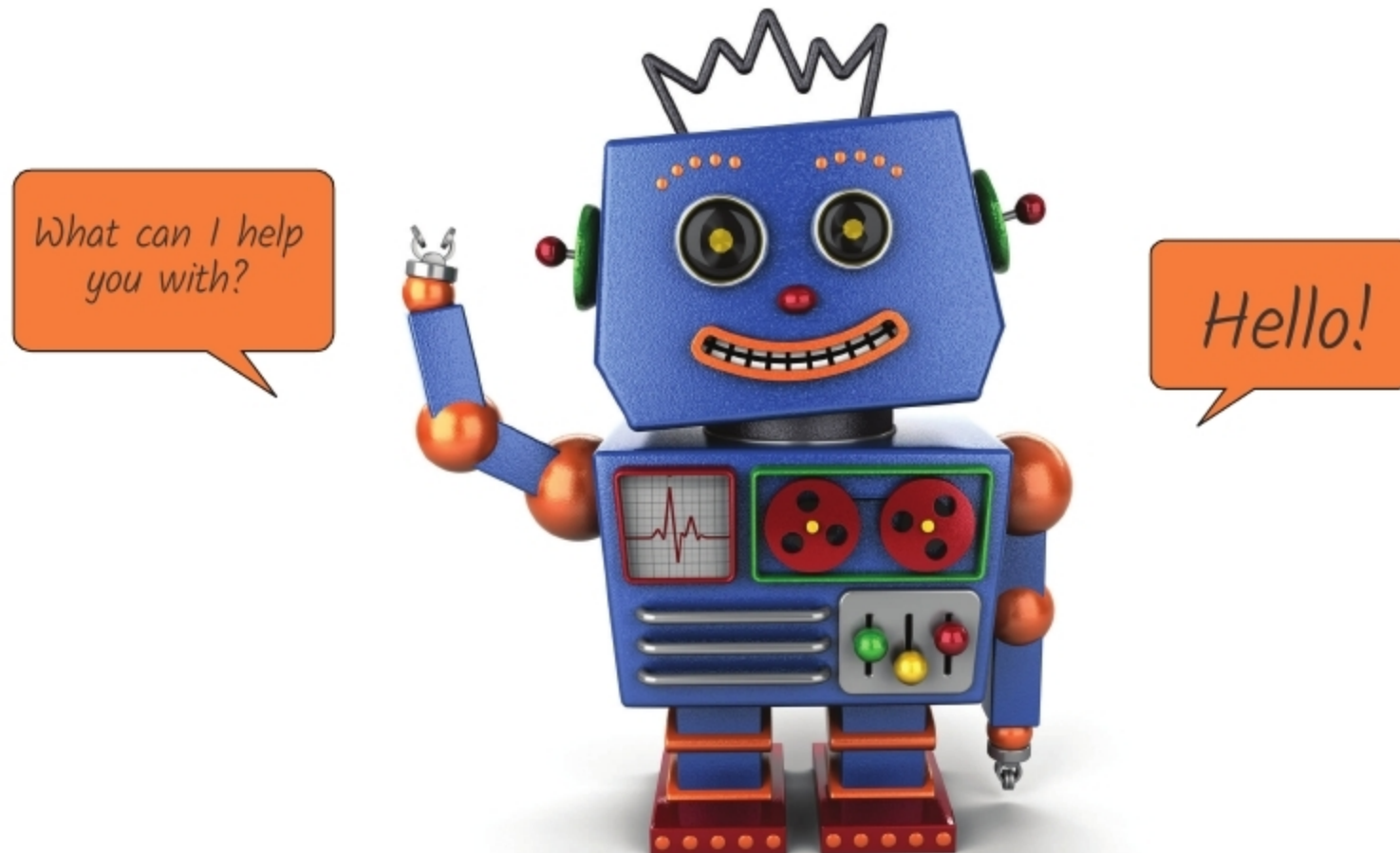


The Connect Between Artificial Intelligence and Chatbots

Automation is taking over many human functions at many levels. One example is of the function of answering queries in various fields like banking, software and e-commerce. Utilising a bot saves the company the cost of employing a person, since the bot will not join another firm, requires no salary and no perquisites, while it works diligently! What more could an entrepreneur want?



Have you ever got confused after hearing terms like artificial intelligence, machine learning, deep learning and data science? Is there any difference between them or do they all mean the same thing? If there are differences, then why do people use them interchangeably?

So, before getting into artificial intelligence and its applications, let's understand a bit about all these terms and see how they are related.

Note: These technologies and methods are evolving dynamically and are very vast in scope. There are different ways to define them and I have tried to explain them in a layman's terms.

Artificial intelligence (AI)

Artificial intelligence is the technique that enables machines (mostly computers) to replicate human behaviour. This is a big umbrella term covering Big Data, machine learning, data engineering, data science and a lot more. There could be a combination of multiple small sub-components to design one solution. Refer Figure 1 for pictorial representation.

Machine learning (ML)

Machine (computer) learning (studying) involves training a machine to learn by its previous experiences. In simple words, ML is a set of tools that helps in learning from data. ML widely uses statistical tools to make sense of given data sets. ML is divided into three categories.

- Supervised learning:** This deals with already known data, which means labelled data sets with a finite number of characteristics—like the data of your employees with their names, ages, salaries and designations.
- Unsupervised learning:** This deals with unknown data or unlabelled data sets with unknown characteristics, like a bucket full of fruits, but you don't know how many varieties there are in it.
- Reinforcement learning:** This allows the machines to automatically determine the ideal behaviour within a specific context. Reinforcement signals (reward and feedback) help a machine to learn how it should behave.

Deep learning (DL)

Deep learning is a branch of machine learning that is based on